

**Highlighted contents are required.*



[Introduction to quantitative biology A]

Term: [Fall]

Credits: [3]

Class Hours: [48] (Teaching Hours: [48], Lab Hours: [0],
Practice Hours: [0])

Course Coordinator

Name	Po-Yi Ho	Contact Info	poyiho@westlake.edu.cn
Office Address	E3-211	Office Hours	

Course Instructor(s) [List all other instructors here. Add more blanks if needed.]

Name #1	Yifan Yang	Name #2	Po-Yi Ho
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Name #3		Name #4	
Contact Info		Contact Info	

Course Description [Shall include course introduction, prerequisites, etc.]

This course introduces researchers from diverse backgrounds to quantitative principles in biological systems. These principles aim to provide for biology a unified and quantitative framework comparable to that of physics or mathematics. We cover several perspectives, including dynamical systems analysis of biological networks and how biological systems represent information in face of stochasticity. Model systems that we analyze include gene regulation, cell biology, neural dynamics, and evolution. Close connection between theory and experiment is emphasized. We also discuss connections to modern data-driven approaches in biology. Problem sets involve the application of mathematical models and computational tools to test candidate principles. The major learning goal is fluency of the conceptual and quantitative tools required to think quantitatively about biological systems.

This course is listed as two parts “Introduction to quantitative biology A/B”. It serves as an entry point for the Quantitative Biology and Complex Systems PhD curriculum. It prepares students for more advanced courses in dynamical systems, stochastic processes, statistical mechanics, and data-driven approaches to biology.

Prerequisites: Calculus, linear algebra, statistics, general physics, general biology

Learning Objectives

Upon completing this course, students will be able to:

- (1) Develop numerical and physical intuition for biological systems.
- (2) Translate biological problems into quantitative models and analyze them versus data.
- (3) Analyze the behavior of biological systems in dynamical and stochastic terms.
- (4) Communicate across disciplinary boundaries.

Learning Resources

Nonlinear Dynamics and Chaos. Steven Strogatz.

An Introduction to Systems Biology. Uri Alon.

Biophysics: Searching for Principles. William Bialek.

Thinking Probabilistically. Ariel Amir.

Physical Biology of the Cell. Rob Phillips, Jane Kondev, Julie Theriot, and Hernan Garcia.

Quantitative Biosciences. Joshua Weitz.

Assessments and Grading [Each criterion can be weighted freely totaling 100%.]

Assessment Criteria	Percentage
Attendance	0%
Participation	20%
Assignment	80%
Course Paper	0%
Exam (open-book/closed-book)	0%
Project presentation	0%

Grade	Assessment Standard
A	90-100 points
B	80-89 points
C	70-79 points
D	60-69 points
F	0-59 points

Course Policies

Course Schedule

Week	Theme/Topic	Instructor(s)	Assignments
1	What are quantitative principles in biology?	Yifan Yang Po-Yi Ho	Pset 1
	Numbers and scales, and how to build simple models	Yifan Yang	
2	Chemical kinetics, the lac operon, and transcription networks	Yifan Yang	
	Nonlinear regulation and solving equations graphically	Yifan Yang	
3	Network motifs and dynamical functions	Yifan Yang	Pset 2
	Positive feedback, bistability, and cellular decision-making	Yifan Yang	
4	Negative feedback, delay, and oscillations	Yifan Yang	
	Excitability, thresholds, and transient cell states	Yifan Yang	
5	Chemotaxis and random walks	Po-Yi Ho	Pset 3
	Chemotaxis and random walks	Po-Yi Ho	
6	Cell cycle regulation and correlations	Po-Yi Ho	
	Morphogenesis and information theory	Po-Yi Ho	
7	Morphogenesis and information theory	Po-Yi Ho	Pset 4 Final project assigned
	Damage dynamics	Yifan Yang	
8	Statistical mechanics description of the lac operon	Yifan Yang	
	Hemoglobin and cooperativity	Yifan Yang	